

ISEF-derived computational biology tools: award-corpus audit, family-held-out toxin benchmark, and AppA model provenance

Current corrected research manuscript | September 25, 2026 | Ten runnable candidate directions are present; this 19-page aggregate is not a final per-project paper. The updated owner floor is 50 pages of substantive research content per project, excluding headings and references; benchmark and external-resource gates remain open.

Abstract. We built a reproducible corpus of 125 CBIO award project-year rows from seven Society for Science listings. A winner-only corpus cannot estimate award odds or validate a selection rule. A five-criterion rubric remained usable as a structured scoring CLI but has not undergone independent inter-rater testing. Of ten derived projects requested, this detailed study tests VenomFunction-2, a sequence classifier benchmark, and PhosGene, an AppA codon and phosphorus-balance sandbox; an updated source-limited inventory below describes the other eight runnable builds without promoting their results to validated discoveries. In 1,814 deduplicated reviewed snake-toxin sequences, random-split mean AUROC was 0.98044, falling to 0.59123 under family-held-out folds. An attribution proxy failed. PhosGene preserved protein translation and mass balance but could not verify a pig phenotype; an evidence validator refuses to compute that phenotype from incomplete in-vivo inputs. We individually fetched and scored 120 named UniProtKB entries. The shared scientific service ledger counts 17 uses across both candidates, below the 40-per-project requirement. These are bounded research results, not clinical or animal claims.

1. Research question and frozen scope

Question A: can official CBIO awards support a dependable design rubric? Question B: does a toxin-keyword predictor generalize outside annotated venom families? Question C: does a computational AppA construct optimizer reproduce a measurable fecal-phosphorus effect? The first is not answerable as an award-probability question from winning titles alone. The second has a falsifiable benchmark with a negative generalization gap. The third is currently unassessable from parameter provenance; a within-project evidence gate now identifies what is missing. These negative and unassessable waypoints are not terminal failures: further within-project pivots must retain them and seek a genuine useful result. We keep these outcomes separate.

The user requested ten new ISEF-derived projects, each with at least 120 accession datasets, 40 external scientific tools, formulas, a falsifiable discovery or broken benchmark, a usable tool, a full paper and delivery. This manuscript gives detailed methods for candidates 1-2 and a separately labeled status inventory for eight newly built candidate directions 3-10. All ten have runnable code or audits, but none is asserted to meet every per-project scientific/resource gate; the status inventory is not a full methods/results paper for each of the eight. Three earlier tool prototypes are retrospective context and do not increase the ten-build count. The item-27 study has 120 individually fetched-and-used toxin entries, 129 separately retrieved AppA-family annotation entries, and 17 counted service uses assigned separately, never pooled to satisfy a candidate gate. Of these services only six are assigned to VenomFunction and eleven to PhosGene; a service must be materially used within the named candidate to count toward that candidate, so no shared-ledger completion claim is made. Shared counts do not automatically satisfy 120/40 for either standalone candidate: VenomFunction has the 120 direct toxin entries but fewer than 40 tools, and PhosGene now has 129 direct entries used in an exploratory AppA-family provenance audit, but not 120 experimentally validated phytase activity measurements, and still fewer than 40 tools or a pig outcome. The nonwinner denominator and three blinded outside raters are absent.

2. Source provenance and reproducibility

Award listings came from the original Society for Science full-awards pages. A project is one unique year/project-code pair after repeated top-award mentions are deduplicated. Each TSV row records year, code, title, award tier and original source URL. 2020 is excluded from this award-listing corpus rather than interpreted as zero CBIO

Source: mega27-27-isef-bioinf-derived-tools | All outcomes limited to cited data and tests

award projects. Source and parser: `survey/data/official_cbio_awards.tsv` and `survey/scripts/collect_awards.py`. The seven pages include <https://www.societyforscience.org/press-release/regeneron-isef-2025-full-awards/> and <https://www.societyforscience.org/press-release/regeneron-isef-full-awards-2023/>.

Year	Awarded CBIO rows
2018	10
2019	13
2021	25
2022	20
2023	20
2024	15
2025	22

The literature and service checks do not manufacture 40 independent sources. One database queried 120 times remains one service. Mirrors of the same record and inaccessible endpoints are excluded. Public, unauthenticated science APIs required no external account login; private Drive delivery uses the connected owner account. Source URLs, test IDs, result paths and the counted judgment for every tried tool are in `external_tools/ledger.tsv`.

3. Winner-only award analysis: what can be measured

The title audit uses fixed literal/regex-like tokens, never semantic coding of the full abstract. It counts explicit wording rather than actual underlying methods. The 125 listing records are neither patient/protein datasets nor a sample of all science-fair submissions. A zero count is compatible with a method used but omitted from the title. A high count cannot establish judges rewarded that technique. The six original “what wins” patterns remain candidate prompts, not validated rules.

Title phrase	Explicit matches in 125 titles
CNN explicitly named	0
deep learning explicitly named	19
graph explicitly named	16
machine learning explicitly named	18
clinical or disease keyword	41
attribution or interpretability explicitly named	0
low cost or access explicitly named	1
NLP explicitly named	0

A prospective award-probability analysis would require submitted projects including nonwinners, consistent category coding, covariates for year and award tier, and an analysis plan fixed before comparing methods. We have no such denominator. A 2023 suicide-prevention example appears outside CBIO and must not be folded into this CBIO award corpus. See `survey/INTERPRETATION.md` for the individual six-pattern audit.

Observed title share = $n(\text{explicit title matches}) / 125$; this is NOT $P(\text{award} \mid \text{method})$.

Citation: Society for Science 2025 full awards, URL above, plus the per-row official URLs in the award TSV. Re-running `survey/scripts/collect_awards.py` gives the source corpus. The parser is independently checked by `tests/test_awards.py` and `tests/test_pattern_audit.py`.

4. Rubric as an instrument, not a validated predictor

The scorer takes five required integer 0-4 fields: real data availability, novelty, feasible scope, measurable verdict and field or clinical value. Total is the unweighted sum; tier thresholds are strong ≥ 17 , promising ≥ 13 , needs work below 13. Missing, extra, Boolean, fractional and out-of-range scores are rejected. This prevents malformed input, not subjective disagreement or judge validity. The anchors are displayed by the CLI and the input remains a human judgment.

$T = d + n + f + m + v$, where each term is an integer in $\{0,1,2,3,4\}$; $0 \leq T \leq 20$.

Source: `mega27-27-isef-bioinf-derived-tools` | All outcomes limited to cited data and tests

tier(T) = strong if $T \geq 17$; promising if $13 \leq T < 17$; needs work otherwise.

Candidate	Total / 20	Weakest dimension
VenomFunction-2	18	novel_angle
DreamMarker	18	measurable_verdict
RelapseCV	18	novel_angle
ConnectoScope	17	clinical_or_field_value
PhosGene	17	real_data_available

Scores were frozen before the two new tests in protocols/prospective_2026-09-25.md. That dated local file is not an independent preregistration service. The earlier three selected builds were retrospective. The prospective two were preselected, unblinded and far too few to validate predictive calibration. The planned independent three-rater study has a protocol and blank template only; it has not recruited or measured anyone. No kappa or alpha estimate exists.

Unweighted exact agreement = number of equal paired scores / number of paired criteria; no value is available without raters.

A proper rater study would anonymize and randomize idea briefs before outcomes are revealed, use three separately consenting qualified scorers, preserve raw integer ratings and report each pair, not cherry-pick the most agreeable pair. The user has not selected recipients or shared-material scope for outside contact. This scientific gap is separate from CLI usability.

5. VenomFunction-2: task, entries, split design

We retrieved 2,068 reviewed Swiss-Prot snake-toxin search rows in five bulk pages. The task label is UniProt Neurotoxin keyword, not measured potency, clinical benefit, receptor binding or an experiment. The family is a UniProt similarity-comment family annotation. Sequences shorter than 25 residues, unnamed-family records and exact duplicates were excluded; one exact-sequence conflicting-label group was removed as ambiguous. The final training/evaluation corpus has 1,814 unique sequences, 542 keyword positives, across 25 named family groups. Bulk IDs alone do not count toward the strict individual-accession gate.

The classifier is a binary character 3-mer CountVectorizer with min_df=2 and balanced logistic regression. Vocabulary is fitted only on the training fold. A prevalence-only dummy baseline is refit per fold. Five random stratified folds are compared with five disjoint family folds; these are different target populations and not interchangeable matched samples. A large random-fold AUROC can arise from homologous motifs shared across train and test; measured 3-mer overlap is audited below.

$$x_s[k] = 1 \text{ if protein sequence } s \text{ contains amino-acid 3-mer } k, \text{ else } 0.$$

$$p(y=1 | x) = 1 / (1 + \exp[-(\beta_0 + \sum_k \beta_k x_k)]).$$

$$L(\beta) = \sum_i w_i [-y_i \log(p_i) - (1-y_i) \log(1-p_i)] + \lambda ||\beta||_2^2.$$

$$\text{AUROC} = P(\text{score}(x_{\text{positive}}) > \text{score}(x_{\text{negative}})) + 0.5 P(\text{tie}).$$

$$\text{AP} = \sum \text{over recall steps } j \text{ of } (\text{Recall}_j - \text{Recall}_{(j-1)}) * \text{Precision}_j.$$

$$\text{Gap} = \text{mean}(\text{random-fold AUROC}) - \text{mean}(\text{disjoint-family-fold AUROC}).$$

5.1 random folds

Fold	Train	Test	Positive test	AUROC	Average precision	Baseline AP
0	1451	363	108	0.98	0.9519	0.2975
1	1451	363	108	0.9854	0.9645	0.2975
2	1451	363	109	0.9787	0.9584	0.3003
3	1451	363	109	0.9879	0.9786	0.3003
4	1452	362	108	0.9703	0.9343	0.2983

Five assessable folds; arithmetic mean AUROC 0.98044. This is not an external independent test cohort. Full per-fold family lists, test counts and metrics reside in venom/data/evaluation.json. No bootstrap confidence interval is claimed here.

5.2 family held out folds

Fold	Train	Test	Positive test	AUROC	Average precision	Baseline AP
0	1306	508	327	0.4962	0.6365	0.6437
1	1459	355	127	0.5127	0.384	0.3577
2	1497	317	22	0.7022	0.1269	0.0694
3	1497	317	46	0.5622	0.154	0.1451
4	1497	317	20	0.6828	0.1051	0.0631

Five assessable folds; arithmetic mean AUROC 0.59123. This is not an external independent test cohort. Full per-fold family lists, test counts and metrics reside in `venom/data/evaluation.json`. No bootstrap confidence interval is claimed here.

5.3 What the split comparison does and does not show

The arithmetic mean AUROC is 0.98044 random versus 0.59123 family-held-out, difference 0.38921. This is a reproducible broken-benchmark finding: random-split keyword performance does not transfer cleanly to annotated unseen protein families under this model and task. Class composition differs by held-out family; the gap is descriptive, not an isolated causal measure of homology leakage. Unmeasured cross-family sequence resemblance and curation bias remain possible. Baseline prevalence is fold-specific, so AP should always be read alongside positive count and baseline AP.

5.4 Follow-up: measured sequence-similarity exposure

After the frozen benchmark, a separate follow-up measured whether test proteins actually resemble proteins available for training. In each original fold, the method forms the set of contiguous three-amino-acid substrings for every test and training sequence, computes pairwise Jaccard similarity, and retains the maximum training match for each test record. Unlike a model score, this overlap has no fitted label coefficients. The local analysis plan was written in the same session just before this computation; it is not independent preregistration, and the original AUROC difference was already known.

$$\text{Jaccard}(s,t) = |\text{three_mers}(s) \cap \text{three_mers}(t)| / |\text{three_mers}(s) \cup \text{three_mers}(t)|.$$

$$\text{Maximum exposure}(s, \text{train_fold}) = \max_{t \in \text{train_fold}} \text{Jaccard}(s,t).$$

Frozen split	n test	Weighted mean max Jaccard	Fraction max Jaccard ≥ 0.80
random	1814	0.6210	0.3352
family_held_out	1814	0.0442	0.0000

Random-fold test proteins have mean max Jaccard 0.6210 to a training protein; 33.5% have a match ≥ 0.80 . Family-held-out test proteins have mean max 0.0442 and none ≥ 0.80 . This documents a large similarity exposure difference, which makes the high random AUROC less persuasive as evidence for a novel-family use case. It does not isolate a causal leakage effect: group composition and label prevalence differ; threshold 0.80 was a fixed descriptive cutoff, not a biological homology definition. Fold-specific metrics and source code are in `venom/data/similarity_leakage.json` and `venom/scripts/similarity_leakage.py`.

5.4 random similarity by original fold

Fold	n test	Mean max Jaccard	Median	10th pct	90th pct	Share ≥ 0.80	AUROC
0	363	0.613	0.639	0.259	0.925	0.322	0.980
1	363	0.609	0.633	0.241	0.928	0.322	0.985
2	363	0.620	0.647	0.246	0.927	0.350	0.979
3	363	0.631	0.675	0.268	0.927	0.336	0.988
4	362	0.632	0.684	0.248	0.931	0.345	0.970

Spearman correlation across only five fold means and five fold AUROCs is -0.40. This small-n fold-level descriptive statistic is not a significance test and is confounded by fold family and class composition; no p-value is claimed.

5.4 family held out similarity by original fold

Fold	n test	Mean max Jaccard	Median	10th pct	90th pct	Share >=0.80	AUROC
0	508	0.039	0.039	0.031	0.048	0.000	0.496
1	355	0.041	0.040	0.033	0.049	0.000	0.513
2	317	0.055	0.048	0.035	0.077	0.000	0.702
3	317	0.045	0.043	0.034	0.051	0.000	0.562
4	317	0.045	0.041	0.033	0.068	0.000	0.683

Spearman correlation across only five fold means and five fold AUROCs is 1.00. This small-n fold-level descriptive statistic is not a significance test and is confounded by fold family and class composition; no p-value is claimed.

6. Individually fetched accession audit

To meet a stricter unit of biological data, venom/scripts/validate_individual.py issued 120 separate direct UniProtKB entry reads. Each saved JSON records accession, sequence and source URL, compared with the bulk snapshot. We selected 60 three-finger toxin and 60 phospholipase A2 records, each group 30 keyword-positive and 30 keyword-negative after deduplication. No held-out accession enters training; individual_validation.py produced a prediction from each individually fetched sequence. The selection is balanced by construction and uses the same UniProt source, so it is not a new natural-prevalence or external experimental cohort.

phospholipase a2: n=60, positives=30, AUROC=0.60667, AP=0.58658, prevalence-only AP=0.500. Do not read the two family scores as a pooled clinical metric.

three-finger toxin: n=60, positives=30, AUROC=0.42667, AP=0.48640, prevalence-only AP=0.500. Do not read the two family scores as a pooled clinical metric.

Per-accession audit inclusion = direct entry GET AND identity/sequence validation AND an actual held-out score.

The 120 one-to-one source records are named in venom/data/individually_validated_120.tsv; the scored probabilities, labels, family and URLs are in venom/data/individual_validation.json. Appendix A tabulates all 120 scores. Three-finger AUROC below 0.5 in the balanced sample is a negative result, not a reason to invert labels after seeing the answer.

6.0 Post-hoc family annotation control

On the same five random folds, a model given only the UniProt similarity-comment family label achieved mean AUROC 0.87929, compared with 0.98044 for the sequence 3-mer model. The family feature is an annotation, not a sequence-derived predictor. The encoder is fit only on each training fold, but both label and feature come from UniProt. This post-hoc control diagnoses annotation structure, not a causal decomposition of the sequence model or a fix for held-out-family failure. Frozen local diagnostic plan, code and results:

protocols/venom_family_annotation_diagnostic_2026-09-25.md, venom/scripts/family_annotation_control.py, venom/data/family_annotation_control.json.

Random fold	Family-only AUROC	Sequence 3-mer AUROC
0	0.8766	0.9800
1	0.8879	0.9854
2	0.8955	0.9787
3	0.8712	0.9879
4	0.8653	0.9703

6.1 Diagnostic threshold error audit

After seeing the held-out probabilities, we calculated a descriptive 0.5 decision-threshold confusion matrix without re-fitting the model or selecting a new threshold. This is post-hoc error characterization and is not a prospectively selected operating point. Ranking metrics can remain above chance even when an uncalibrated 0.5

threshold produces zero true positives. Source: results/transfer_error_audit.json; the script recomputes each number directly from the 120 original prediction rows.

Held-out family	TN	FP	FN	TP	Brier	Largest score
phospholipase a2	30	0	30	0	0.371	0.458
three-finger toxin	30	0	30	0	0.392	0.405

The max probability is below 0.5 in both groups. Every 30 keyword-positive entries per family is a false negative at that threshold; specificity looks perfect only because the classifier always predicts negative in this balanced audit. Brier score is a squared-probability error summary and should be compared only within the same label definition and sample selection. The failure may reflect a shifted intercept under family exclusion, altered class mix or insufficient family-transfer signal. The available data cannot distinguish these causes, and post-hoc threshold tuning on these samples would defeat the held-out audit. A next within-project pivot is to freeze a separate nonoverlapping validation-family set, then calibrate a threshold on those families only and evaluate on untouched test families; the current two groups remain a frozen negative, not refitted success.

$$\text{Brier} = (1/n) * \sum_i (\text{predicted_probability}_i - \text{keyword_label}_i)^2.$$

$$\text{Sensitivity at threshold 0.5} = \text{TP} / (\text{TP} + \text{FN}) = 0 \text{ in both 60-entry family samples.}$$

6.2 Threshold transfer without target-label tuning: another negative

A separate exploratory attempt asked whether a decision threshold learned outside both target families could recover positives without reusing target labels. The model was trained on 951 sequences from 23 other families. Five GroupKFold out-of-fold predictions over those training families supplied a single scalar threshold that maximized balanced accuracy. The previously scored 120 direct-entry target proteins were not used to choose this threshold. However, the target outcomes were already visible before this experiment was planned, so it is not independent prospective validation.

The selected training-family threshold was 0.001159, with training-family out-of-fold balanced accuracy 0.5227. It is far below 0.5, already warning of severe distribution shift. On each of the two balanced target groups, the original 0.5 threshold gave 30 true negatives and 30 false negatives. The non-target-calibrated threshold gave 30 true positives and 30 false positives. Both achieve balanced accuracy 0.5 by predicting one class for all records. Thus this threshold transfer did not repair the actual decision problem. It cannot be reselected on target labels after seeing this result without sacrificing holdout integrity.

Family	Original TN/FP/FN/TP	New TN/FP/FN/TP	New balanced accuracy
phospholipase a2	30/0/30/0	0/30/0/30	0.500
three-finger toxin	30/0/30/0	0/30/0/30	0.500

This jointly excluded both target families during model fitting, whereas the original direct-entry scoring excluded one family at a time. Its rankings/AUCs are therefore not apples-to-apples with the earlier AUCs, even though the same target accessions were used. Code and complete result: venom/scripts/transfer_calibration.py and venom/data/transfer_calibration.json. The within-project direction remains open; an independent biological label or a truly untouched external cohort would be needed for a stronger claim.

6.3 Within-project label-coherence check

A second VenomFunction pivot checks a narrow data-quality explanation without redefining the original held-out negative. Registered thresholds 1.00 and 0.95 positional identity before this scan; only equal-length raw direct-entry sequences are comparable by this proxy. The scanner reads the same 120 saved direct-UniProt entry JSON files and lists accession pairs with their keyword, source URLs, family and identity. This is not a new cohort or an optimized classifier.

Across 120 direct entries, 277 pairs had equal lengths; 21 had at least 95% positional identity, and 0 of those had discordant keyword labels. At exact identity, 0 pairs were found. Thus the high-identity equal-length subset does not reveal a direct internal keyword contradiction in this sample. Unequal-length sequences were not aligned; other annotations, nearly identical proteins outside this 120-entry selection, and true functional outcomes remain untested. File: venom/data/label_conflict_audit.json, method: venom/scripts/label_conflicts.py.

Positional identity(s,t) = $\sum_i 1[s_i=t_i] / \text{length}(s)$, defined here only if $\text{length}(s)=\text{length}(t)$.

The original family-transfer result remains negative even after this data-quality check. A curator could use the scanner on later direct-entry files to flag contradictions rather than quietly choosing one label; a zero here does not establish data cleanliness or potency. This tool adds no new external scientific service or accession count to the project gates.

7. Residue-attribution proxy: retained negative

The pre-specified site test replaces a curated active-site position with X and compares the absolute model-logit change with a within-protein control position more than five residues away. This is an input-sensitivity proxy. Catalytic active sites can belong to non-neurotoxin enzymes, so a failure here is not a definitive absence of a neurotoxin causal residue. It cannot prove binding, treatment benefit or counterfactual protein behavior.

$$D_{\text{site}} = |\text{logit } f(\text{sequence with site} \rightarrow X) - \text{logit } f(\text{original sequence})|.$$

$$R_{\text{site/control}} = \text{mean}(D_{\text{site}}) / \text{mean}(D_{\text{matched_control}}).$$

The audit covers 568 protein records and 1103 site-control pairs. Site mean absolute delta was 0.10132, control mean 0.15047, ratio 0.67338; 40.53% of sites exceeded controls. Predicted enrichment failed in this proxy. We retain the negative rather than retune sites after results. The complete positions, provenance and limitations are in `venom/data/active_site_audit.json`.

8. Distinct motif and structure context: annotations, not potency

ScanProsite direct scans found SNAKE_TOXIN PS00272 on P01400 and phospholipase A2 His/Asp signatures PS00118 and PS00119 on P00604. This corroborates broad family identity but not potency. HMMER phmmer queried the P01400 mature sequence against its PDB protein set and returned a *Bungarus candidus* weak-toxin homolog 2jqp_A (E=1.67e-28). PDBe maps that homolog accession Q8AY51 to a 65-residue solution-NMR structure at 100% coverage; RCSB confirms the same PDB chain. It does not show P01400 has a resolved structure. Shared PDB data from PDBe and RCSB count as one structural evidence source.

Exact live-returned URLs and records: `external_tools/prosite_check.json`, `hmmer_check.json` and `structure_check.json`. The RCSB UniProt text-search attempt returned HTTP 400, and an earlier PDBe P00604 mapping returned 404; those attempts were not counted. Motif, homology and structure are orthogonal checks of identity or fold availability, not a new potency assay or a validation of the predictor.

9. PhosGene: coding-sequence arithmetic and physiology gap

GenBank U00096.3 resolves the AppA CDS and translation corresponding to UniProt P07102. The older M58708 CDS has an ambiguous base at one codon; the K-12 genome provides a resolved codon with the same translated amino acid. The Kazusa taxon 83333 codon table has 64 codon entries based on only 14 coding sequences and 5,122 codons, so its reference weights are sparse. The optimizer replaces each amino acid codon with the highest-frequency synonymous codon and retains a terminal stop. It checks translation identity and input bounds. No construct was synthesized or tested for host expression.

$$w(c | a) = \text{observed_count}(c) / \max_{\{ \text{synonymous } c^* \text{ encoding } a \}} \text{observed_count}(c^*).$$

$$\text{CAI}(\text{sequence}) = \exp[(1/L) \sum_{i=1}^L \log w(c_i | \text{amino_acid}_i)].$$

Observed AppA CAI is 0.610734 against the sparse table; the unrestricted maximum is 1.0 by construction. A ceiling reached by replacing all codons is an arithmetic test, not evidence for increased expression or secretion. GC-window constraints, repeat filtering and host-specific secretory regulation are not supported in the current optimizer.

The enzyme sandbox takes substrate mmol/kg, enzyme units/kg, Km mM, fluid volume L/kg, duration min and a bounded pH factor. It Euler-integrates a Michaelis-Menten-like step with a nonnegative substrate bound. A

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literature Km of 0.79 mM is from an in-vitro AppA assay; assay units do not transfer automatically to pig intestinal residence or active concentration. Phytate hydrolysis and newly absorbed phosphate are different quantities.

$$\begin{aligned}
 S_{\text{mM}} &= S_{\text{mmol_per_kg}} / V_{\text{L_per_kg}}. \\
 \text{rate_mmol_per_kg_min} &= (\text{enzyme_units_per_kg}/1000) * \text{pH_factor} * S_{\text{mM}} / (\text{Km_mM} + S_{\text{mM}}). \\
 S(t+dt) &= \max(0, S(t) - \min(\text{rate} * dt, S(t))). \\
 \text{new_retained_P} &= \min(\text{baseline_fecal_P}, \text{dietary_P} * \text{phytate_fraction} * \text{hydrolysis_fraction} * \text{absorbed_fraction}). \\
 \text{predicted_fecal_reduction} &= \text{new_retained_P} / \text{baseline_fecal_P}, \text{ if } \text{baseline_fecal_P} > 0.
 \end{aligned}$$

The compartment mass balance bounds hydrolyzed substrate between zero and initial substrate; the fecal-P balance caps retention at baseline fecal P. These are invariants of a user-supplied model, not measurements. Missing in-vivo enzyme rate, pH history, gut fluid volume, residence time and absorption coefficients make an independent fecal-P phenotype comparison unassessable. The older 54.1% prototype estimate is not independently reproduced here.

9.1 Reaction identity and source disagreement

KEGG eco:b0980 maps appA to a 4-phytase/acid phosphatase context, while b4055 maps another phosphatase AphA. UniProt P07102 assigns an unspecific EC and one positional pentakisphosphate product. ExPASy ENZYME and Rhea list generic enzyme-class reactions and can use different inositol phosphate positional conventions. It would be incorrect to use any one generic EC step as proof of six phosphate-release steps in vivo. All source claims and exact URLs are in `external_tools/phos_reaction_check.json`.

BRENDA Km entries span assay contexts, not validated pig luminal conditions. STRING resolves AppA to the b0980 gene neighborhood without showing dietary phenotype. EcoCyc independently links AppA monomer to appA/b0980 and periplasmic location; QuickGO returns 21 evidence-coded annotations including IDA/EXP activity/localization, often citing EcoCyc or UniProt. IntAct returns three partner rows with two interaction AC values from one culture pull-down study, not independent animal assays. PRIDE Archive protein P07102 search maps to three project identifiers, but direct project metadata lists horse/mouse/bovine, human and C. elegans, no pig context. That mapping is not itself evidence of organism-specific AppA peptide assignment. These checks locate a gene and a function-context disagreement rather than filling the kinetic gap. Source files:

`external_tools/taxonomy_and_brenda.json`, `string_check.json`, `appa_annotations.json`, `intact_check.json`, `pride_check.json`.

The 2018 multi-enzyme pigs study at <https://elifesciences.org/articles/34286> reports fecal P lower by 45.8% under a corn-soybean diet and 34.8% under a wheat/corn/soybean/bran diet (its abstract pools N and P into a 23.2%-45.8% range; the 23.2% lower endpoint is fecal N), whereas the original Enviropig report at <https://europepmc.org/article/MED/11479566> reports up to 75%. These are different systems and comparisons. A pooled 20%-60% band is not a defensible prespecified verification target. No pig-gut measurement has been generated by this repository. The next within-PhosGene direction is a parameter-identifiability tool: report the entire feasible fecal-P outcome range over evidence-bounded inputs, and leave unknown dimensions visibly unbounded instead of selecting values to hit a published pig outcome. This is a planned pivot, not a completed scientific result.

9.2 Within-project pivot: provenance-gated phenotype identifiability

Measured-variant eligibility check. A 2023 AppA engineering paper reports six *P. pastoris* parent/mutant culture activity rows with matched mutation identities (<https://www.frontiersin.org/journals/bioengineering-and-biotechnology/articles/10.3389/fbioe.2023.1167530/full>). An exact table transcription and tested comparison in `phosgene/benchmarks/frontiers_appa_variants.tsv` and `frontiers_appa_audit.json` finds the published D7 mutant at 837.2 ± 17.8 U/mL versus its own parent 508.6 ± 21.5 U/mL (ratio 1.6461). These prior investigators measured this positive result, not PhosGene. Their parent is reported 97% homologous to PDB 1DKP; exact sequence/numbering identity with our P07102/U00096 reference is unverified. Culture activity, codon adaptation

and pig fecal P are different endpoints, so this cross-check neither wins an algorithm benchmark nor validates an animal claim; published literature is not counted as an external research tool.

The unresolved pig-phenotype reproduction led to a concrete within-PhosGene pivot rather than an invented positive effect. The new identifiability validator demands a numerical value, unit, source URL, organism, assay or feeding condition, and transfer rationale for every parameter used to turn AppA activity into a predicted fecal-P effect. Its requirement table is fixed in protocols/phosgene_identifiability_2026-09-25.md, and the implementation is phosgene/identifiability.py. A non-pig assay cannot silently stand in for pig-lumen physiology: the provided rationale must be substantial, but a substantial statement alone would not scientifically validate transfer. Owner/source review would still be required before interpreting a calculation as an animal result.

The actual source packet contains an in-vitro purified E. coli AppA Km of 0.79 mM with PMID 10696472 and explicitly blank pig-transfer rationale. No pig-gut volume, active enzyme exposure, residence time, pH activity history, diet partition, baseline fecal fraction or absorption fraction is supplied. The validator returns 10 gaps, status unassessable and phenotype null. This failure is the intended scientific guardrail. A synthetic all-fields packet is used only by a unit test to check input bounds and mass-balance arithmetic; its output is never the reported pig phenotype. Committed result: phosgene/data/identifiability_audit.json.

Required source-grounded dimension	Expected unit	Current verdict
substrate_mmol_per_kg	mmol/kg	missing or unsupported transfer
enzyme_u_per_kg	U/kg	missing or unsupported transfer
km_mmolar	mM	missing or unsupported transfer
fluid_l_per_kg	L/kg	missing or unsupported transfer
duration_min	min	missing or unsupported transfer
ph_factor	fraction	missing or unsupported transfer
dietary_p_mmol	mmol/kg	missing or unsupported transfer
baseline_excreted_fraction	fraction	missing or unsupported transfer
phytate_fraction	fraction	missing or unsupported transfer
absorbed_fraction	fraction	missing or unsupported transfer

This pivot yields a usable decision aid: researchers can see which original measurements are needed for a falsifiable pig comparison. It is a method/validation result, not a new biological effect size or proof the codon optimizer improves expression. The tool itself is local code, so it adds zero external tools and zero individual accessions to the strict gates. The original negative waypoint remains visible beside this narrower useful deliverable.

9.3 Exploratory AppA-family provenance and transfer audit

To make better use of public sequence evidence without pretending it supplies missing pig physiology, a follow-up queried reviewed UniProtKB entries carrying InterPro IPR000560, a family cross-reference verified on the direct AppA P07102 entry. The query returned 89 named entries; all 89 were retrieved at separate accession JSON endpoints, checked for exact primary accession, reviewed status and family cross-reference, and used in an untuned amino-acid 3-mer Jaccard comparison with P07102. Each record preserves source URL, SHA-256, sequence, organism, name, EC evidence and the explicit refusal to transfer its rate to pig gut. This is an exploratory source-scope study; the local plan preceded entry fetching but followed the original pig-model failure.

Only 16 of the 89 direct records are explicitly named phytases. 34 carry at least one EC annotation with an experimental PubMed citation, which is still not a common-substrate or pig-lumen assay. The maximum non-anchor 3-mer Jaccard with AppA is 0.07798. A broad family match, name or low descriptive k-mer overlap does not establish equivalent catalytic products, secretion or in-vivo rate; these entries cannot calibrate the fecal-P sandbox. A second, pre-stated unreviewed bacterial phytase-label stratum added 40 disjoint direct entries. All 40 carry explicit unreviewed phytase names, but zero experimental EC PubMed citations; maximum AppA 3-mer Jaccard is 0.13043. The combined narrow family-provenance audit uses 129 direct-fetched protein records (89 reviewed, 40 unreviewed), which meets an accession-count condition for that annotation tool only. It does not represent 120 experimentally validated AppA-like kinetic measurements and adds no distinct service beyond

already counted UniProtKB. The 10-gap phenotype audit remains null. Full result:
 phosgene/data/hap_family_audit.json, method: phosgene/scripts/hap_family_audit.py.

10. Calibration and failure modes

A Beta(1,1) prior with three selected retrospective runnable builds produces Beta(4,1), posterior mean 0.800 and equal-tail interval [0.398,0.994]. Counting the two newer runnable candidates would yield Beta(6,1), mean 0.857 and interval [0.541,0.996] for the narrowly defined end point of runnable construction. These values are not a biological success rate, a probability of winning an award, or evidence the rubric predicts discovery. An unassessable animal outcome cannot be relabeled a success just because a CLI runs.

$$P(q | s, n) = \text{Beta}(1+s, 1+n-s) \text{ under a uniform Beta}(1,1) \text{ prior.}$$

$$E[q | s, n] = (1+s)/(2+n); \text{ with } s=n=3, E[q]=4/5.$$

Selection of high-scored ideas inflates apparent build completion. With only two new post-protocol tests and no blinding, weakest-criterion calibration is not statistically assessable. The independent rater template is blank. VenomFunction-2 shows a useful failure of a homolog-sensitive split, while PhosGene fails a source-parameter sufficiency test. Both are legitimate bounded outcomes with different scopes.

11. External tool audit and still-open conditions

The strict ledger has 24 rows documenting service uses and attempted calls (some repeated providers), 17 counted across the two detailed candidates, but this aggregate is not a candidate-level gate. VenomFunction has 6/40 (short by 34); PhosGene has 11/40 (short by 29). Each counted service has a source URL, preregistered or clearly exploratory test ID, output file and interpretive use. Exclusions include the official award page as a corpus rather than an analysis tool, a Pfam endpoint hosted by InterPro as the same service, a failed PDBe request, failed SABIO-RK, failed RCSB search, RCSB mirror of a PDB record, and NCBI BLAST still waiting. This accounting is more conservative than a list of URLs or software names.

Scientific service or attempt	Arm/test	Counted?	Meaning of check
Society for Science	award survey A1	0	Official award evidence source, not a scientific analysis tool; seven pages
UniProtKB	VenomFunction V0	1	120 direct-fetched entries actually scored, one service
InterPro	VenomFunction V1	1	IPR003571 family cross-check
AlphaFold DB	VenomFunction V5	1	Predicted model confidence, not function
NCBI Entrez/GenBank	PhosGene P0	1	AppA CDS; NCBI taxonomy same provider not counted separately
Kazusa Codon Usage Database	PhosGene P0	1	64-codon frequency table in optimizer
KEGG	PhosGene P2	1	AppA gene identity versus wrong AphA gene
ExPASy ENZYME	PhosGene P3	1	4-phytase class reaction/isomer
Rhea	PhosGene P4	1	Reaction isomer disagreement against UniProt direct annotation
BRENDA	PhosGene P1	1	Heterogeneous Km contexts fail to identify in-vivo model parameters
STRING	PhosGene P8	1	P07102 maps to b0980 AppA, periplasmic annotation
InterPro-hosted Pfam	VenomFunction V2	0	Same InterPro service/member signature; not independent
PDBe	VenomFunction V3	0	404 no data
SABIO-RK	PhosGene P6	0	Redirected UI 404; no kinetic result

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ExPASy ScanProsite	VenomFunction V8	1	Motif scanning of two named exemplars, distinct analytical service from ExP
EBI HMMER phmmer	VenomFunction V9	1	P01400 PDB homology result, top weak-toxin neighbor is not an activity meas
RCSB PDB Search	VenomFunction V11	0	P01400 UniProt exact-match attributes rejected HTTP 400; no structural resu
PDBe	VenomFunction V12	1	Prior HMMER homolog Q8AY51 has NMR chain 2jqp_A with mature-sequence covera
RCSB PDB Data	VenomFunction V12	0	Confirmed same PDB entry and chain; shared underlying structural record, no
EcoCyc BioCyc	PhosGene P9	1	Exploratory gene-to-protein mapping b0980 and curated periplasmic context
EBI QuickGO	PhosGene P10	1	Exploratory evidence-code provenance audit of GO localization and phytase;
NCBI BLASTP	VenomFunction V10	0	Submitted P01400 Swiss-Prot search, RID WAITING; NCBI provider collapsed wi
EBI IntAct	PhosGene P12	1	Culture pull-down partner rows; two AC values, same publication, no in-vivo
PRIDE Archive	PhosGene P15	1	Exploratory protein-to-project mapping and sample-organism audit: 3 project

These checks contribute to annotation/identity/reproducibility and not all measure biological function. No fabricated count is applied to GitHub, Drive, shell, PDF renderer, browser, web search or repeated requests to one endpoint. The full exact URLs and paths remain in external_tools/ledger.tsv and per-source result files. The tool gate remains open. A failed or slow service is logged as such and not silently replaced with an invented successful query.

12. Reproduction and responsible claims

Install project dependencies, then run `PYTHONPATH=src:. python -m pytest -q`. Rebuild award list with `python survey/scripts/collect_awards.py`, toxin bulk source with `python venom/scripts/fetch_uniprot.py`, evaluation with `OPENBLAS_NUM_THREADS=1 python venom/scripts/evaluate.py`, active-site test with `PYTHONPATH=src:. python -m venom.scripts.audit_active_sites`, and direct-entry audit with `PYTHONPATH=src:. python -m venom.scripts.validate_individual` followed by `python -m venom.scripts.individual_validation`. Fetch AppA and codon reference with `python phosgene/scripts/fetch_reference.py` and run `PYTHONPATH=src:. python -m phosgene`. Live endpoints can evolve, so committed result JSON and dated URLs are the result provenance; re-runs must compare source IDs, labels and sequences, not assume unchanged annotations.

Safety and scope: toxin analyses compare public sequences and database keywords and do not supply a biological construction protocol or claim clinical deployment. The codon optimization is a computational arithmetic demonstration, with no synthesized construct, animal study, medical recommendation, or environmental release. The result is a usable benchmark/rubric/sandbox at their stated bounded scopes, not world-best potency prediction or a validated pig system.

Remaining for user request: enough genuinely distinct tools to meet 40 per candidate, not merely 24 for a shared ledger; an activity/phenotype-grounded use of PhosGene accessions beyond its completed 129-entry annotation audit, and separate biological assay or accession validation appropriate to each later build; an independently grounded phenotype comparison; three blinded outside raters with owner-approved recipients; candidate builds 3-

Source: mega27-27-isef-bioinf-derived-tools | All outcomes limited to cited data and tests

10 requiring separate full scientific papers, 120+ individually used biological datasets and 40+ distinct scientific tools per project; stronger independent biological benchmarks and external validation where their current gates fail; this program manuscript is a substantive working paper, not eight substituted full papers; release/delivery audit once the scientific gates are actually met. Do not represent ten runnable builds as ten completed scientific projects.

13. Candidate builds 3-10: source-limited program inventory

After the detailed two-project analysis was prepared, separate lanes added eight runnable candidate directions. The entries below summarize their committed READMEs, result objects and methods, not an independent source reanalysis by this manuscript author. They are concise working builds with documented missing inputs, not eight fully validated discoveries or eight full-length papers. Their accession/service gates must be audited separately rather than borrowing the VenomFunction/PhosGene ledger; code, catalog row counts or measured pairs are not automatically 120 individually fetched-and-used biological accessions or 40 independent services.

# / build	Observed bounded result	Unmet biology or resource gate
3 ToxinAudit-LabelScope	120 direct reviewed UniProt entries; 41 target-EC direct citations; AUROC refused	No measured negatives or homolog-disjoint benchmark; 4/40 services
4 CodonContext	Synonymous CDS guard; 120 proteins matched to 120 CDS features in one RefSeq genome	Exploratory PSI expression baseline AUC 0.659; no synonymous-pair yield assay
5 GraphHoldout	CTD/PubMed time split: 114 novel later pairs, only 8 reachable; asthma 2/2 top-10	Tiny reachable set, one retrospective source, most targets cold start
6 TranscriptShift	GEO BAMSE-to-INMA AUC 0.457, reverse 0.521; separate site-cohort audit	Transfer gates fail; cohorts from same papers, no independent hospital claim
7 MicrobeTransfer	120 BioSample AST entries; second 1,178-row six-site metadata audit	Sequence linkage/breakpoints unverified; no hospital-disjoint genome predictor
8 PhageHostScope	38,435 measured pairs; joint holdout AUC 0.709; BASEL lysis audit	No verified Paris-BASEL phage/host join; no external transfer AUC
9 RNAEvidence	Historical comparator AUC 0.647; CLASH/eCLIP leakage audit	858/954 eCLIP rows share gene clusters; no gene-pair validation
10 SpatialArtifactGuard	15,611 aligned spots; 37px image-shift control lowers all six AUCs	Same atlas, no external site; rare artifact folds

The distinct scientific provider ledger for candidate builds 7-10, builds/service_ledger.tsv, counts #7 4/40, #8 3/40, #9 2/40, and #10 1/40. These provider uses are separate from candidate builds 1-6, and the Zenodo files, NCBI sibling endpoints and computational libraries are not additional providers. Candidate #10 directly audited the full gene matrix against all 15,611 metadata/image-position barcodes and counts; its missing check is external-site/pixel registration and a validated patient-level classifier, not absence of a downloaded matrix.

The newly committed per-build ledger for candidates 3-6, projects/evidence_ledger.json and projects/EVIDENCE_LEDGER.md, counts 5/40 services and 120 direct protein entries for #3; 5/40 services and 120 individually retrieved protein entries paired with CDS features from one fetched genome for #4; 3/40 services and 120 bibliographic PMID responses, not biological assay accessions, for #5; 1/40 service and 120 GSM metadata responses, not separately fetched expression datasets, for #6. These are per-build numbers, not additive program closure. Candidate #7 has 120 individually read BioSample records containing AST annotations, but no genome-wise fetched-and-used predictor set or multi-hospital R/S cohort; treating its BioSample retrieval as a completed biological accession gate would erase the stated project requirement. #8 has 97 source phage FASTAs and 105 external-cohort phage FASTAs, but they are different species/source contexts and not a common 120-accession model validation. The other bulk catalog and patient-spot rows are not individually fetched accession datasets.

Candidate 3 provenance: projects/toxin_audit/README.md and results.json (direct UniProt source IDs and EC evidence); 4: projects/codon_context/README.md and synthetic example; 5: projects/graph_holdout/README.md and CTD/PubMed manifest; 6: projects/transcript_shift/README.md and GEO source manifest. Candidate 7: builds/microbe_transfer/README.md and results.json (CARD catalog); 8:

builds/phage_host_scope/README.md and results.json (Paris et al. source); 9: builds/rna_evidence/README.md and results.json (historical miRTarBase); 10: builds/spatial_artifact_guard/README.md and results.json (Zenodo spatial atlas). Each individual README or manifest gives source URLs, methods, test commands and caveats; a reader must inspect those artifacts before relying on a candidate-level claim.

Build-time thresholds and directions are not independent preregistrations. Negative, unassessable and exploratory statuses are retained, and no shared resource count can close the ten separate 120/40 gates. A follow-on program synthesis should independently reproduce the eight builds, audit original biological sources, expand their methods and outcomes into their own full papers, and replace this bounded inventory only after that review.

13.1 Why the eight build verdicts differ

ToxinAudit-LabelScope (#3) retains its original 26-entry panel and expands to 120 separately fetched reviewed UniProt entries under two fixed EC queries. Forty-one expanded entries have direct target-EC experimental citations; the earlier panel had twelve. The absence of such a citation in the others cannot serve as a negative functional assay: an unmeasured protein may still catalyze the reaction. The classifier refuses to fit until independent measured negatives and homolog-disjoint controls exist. QuickGO, InterPro and ChEMBL add contextual annotation and target identity; UniRef50 maps all 120 selected entries to 73 half-identity clusters, 19 containing multiple panel entries. These checks are not biochemical null results or assay-disjoint folds. This is a useful failure-before-modeling control, but no predictive AUROC or novel enzyme-function finding follows. Source: projects/toxin_audit/accessions.json, results.json and external_checks.json.

CodonContext (#4) takes a candidate coding sequence and checks translation preservation, reading frame, terminal stop, GC/motif and window constraints. A first-120 E. coli panel separately retrieves 120 UniProt proteins, mapping them to 120 translated CDS features from a single fetched RefSeq chromosome; it is not 120 separately fetched nucleotide accessions. A PSI protein-expression dataset supports an exploratory accession-prefix split with AUC 0.659, but is not a same-protein synonymous construct test. Angov et al. 2008 provides five paired wild-type/codon-harmonized yield rows, without full paired CDS sequences or comparable CAI; it checks experiment feasibility, not a held-out CAI-gain benchmark. The separate published NAR 2023 mRFP workbook supplies 1,459 sequence/fluorescence rows for one protein and one host, crosschecked against a downstream CSV as one evidence chain. Its exploratory first-eight-codon fixed feature had a held-out within-study Pearson r of 0.722 versus 0.521 for GC features, but a post-result source-header holdout had macro R^2 -0.305 and two negative folds. Those are not 1,459 independent datasets, a matched published benchmark or a new biological discovery. The panel does not supply paired constructs measured in an expression host. Its CAI arithmetic therefore cannot answer whether synonymous changes increase yield, even if a constraint checker can reject impossible or unwanted sequences. This is distinct from PhosGene despite shared P07102 context and must not count that overlap as an independent expression experiment. Source: projects/codon_context/README.md and evidence_ledger.json.

GraphHoldout (#5) uses 556 curated CTD gene-disease-PMID evidence rows with PubMed publication dates. It represents an edge and its reverse as one canonical pair and excludes shared paper IDs across the retrospective time split. Of 114 later novel associations, 106 involve genes absent from the earlier graph; only eight positives are rankable at all. Asthma 2/2 among the eight cannot validate a general ranker when the other disease groups yield 0/5 and 0/1. The 2026 CTD snapshot is filtered by article date, not historical curation date; current Open Targets context can overlap the same papers. The observed cold-start limit and leakage refusal are results; prospective gene-disease prediction is not. Source: projects/graph_holdout/data/results.json and REPORT.md.

TranscriptShift (#6) downloads expression matrices for two GEO cohort accessions, uses a fixed 1,000-probe linear model and trains on an entire study before testing on the other. AUCs of 0.457 and 0.521 fail its two-direction gate. An independent publication supplies GSE110551 with a different platform and explicit site metadata. The old fitted probe model cannot be applied as if probes were harmonized; a separate fixed baseline gives Wroclaw-to-Zurich AUC 0.613 and reverse 0.545. Conditional bootstrap AUC intervals for the original directions are 0.350-0.560 and 0.448-0.596; they do not cover future hospitals or clinical phenotype adjudication.

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This is a paired negative transfer story, not a validated asthma diagnostic. Source: projects/transcript_shift/data/results.json, external_site_results.json and uncertainty.json.

MicrobeTransfer (#7) begins from a 6,463-row resistance-gene index, not an isolate-level susceptible/resistant trial. The follow-up fetches 120 BioSample XML records by exact identifiers and records 442 explicit R/S assay rows. All 30 isolates with interpreted R/S share one collector, so a hospital-disjoint test set does not exist. Assembly links for 29 of these isolates are not genome sequences fetched and used in a predictor. The input contract and abstention rule are operational software, but no clinical susceptibility performance is measured. Matching gene annotations to AST labels without isolate/drug/breakpoint provenance would falsely promote catalog entries into phenotypes. Source: builds/microbe_transfer/ncbi_ast_results.json and README.md.

PhageHostScope (#8) starts from measured lytic host-range pairs, making its data unit different from a docking score. It uses source-provided host and phage trees, holds out groups on both axes and quarantines pairs crossing only one axis. Its within-study AUC 0.709 on 1,120 jointly held-out pairs narrowly crosses an exploratory threshold, but those trees, aliases and group cutoffs were not independently reconstructed or externally confirmed. A separate Klebsiella cohort of 12,970 measured pairs cannot simply accept the E. coli host-tree distance model; a BASEL E. coli source adds 396 measured lysis records but no verified phage/host identity join to the Paris training pairs; absent comparable host representation, the external AUC is intentionally refused. The result is a bounded same-study benchmark, not cross-species host-range prediction. Source: builds/phage_host_scope/results.json and external_results.json.

RNAEvidence (#9) separates curated record grades and publication IDs in a historical 2015 miRTarBase snapshot, quarantining repeated miRNA-gene pairs. A simple assay-propensity comparator yields AUC 0.647 on 454 same-database records, but tested nonfunctional records are selected and cannot represent the unbiased space of all true negatives. The independent miRBench CLASH/eCLIP files instead test a fixed 7-mer sequence rule at the site level; 858 of 954 eCLIP test rows share gene clusters with training context, and two exact site-pair overlaps were quarantined; that is a different endpoint, not external validation of the historical curated gene-pair model. Its risk is endpoint substitution, mirrored literature and constructed negative sites. Neither AUC is a prospective target-interaction ranking result. Source: builds/rna_evidence/results.json, README.md and mirbench_results.json.

SpatialArtifactGuard (#10) checks all 15,611 spot barcodes and metadata counts against a downloaded spatial gene matrix and image-position list, then evaluates a patient-out RNA-QC/center-pixel color baseline. That matrix audit establishes row linkage and count agreement, not precise histology pixel registration or an external site. A pre-stated 37-pixel image-coordinate shift control lowered all six AUCs, supporting a spatial-registration dependence within the same atlas but not an external-site gate. The apparent fold AUCs include groups with only four or one artifact spot; averaging such unstable folds would hide denominator fragility. The atlas is one source across six slides and labels can reflect local annotation practices. The scientific gate remains external-site validation with enough artifacts per held-out patient, not a spot-level random split. Source: builds/spatial_artifact_guard/gene_alignment_results.json, results.json and README.md.

13.2 Cross-candidate interpretation

The eight software builds fail in materially different ways. Some refuse a prediction because the necessary endpoint or linkage was never established (#3, #4, #7); some run a fixed benchmark and preserve poor transfer or limited coverage (#5, #6, #9, #10); #8 has a within-study score but no comparable external test. A negative AUC and a missing validation cohort should not be merged into one failed-performance category. The separate service ledger uses data providers per build, not files or Python packages: #3-#6 count 5, 5, 3, 1; #7-#10 now count 4, 3, 2, 1. No candidate may borrow another candidate accession cohort to claim 120, and none approaches 40 services. The program result is a transparent set of runnable audits and demonstrated evaluation traps, not ten established biological discoveries.

Supplementary material: exact accession predictions, the full award listing, and endpoint-level service provenance are in MEGA27-27-research-supplement.pdf. The main paper stands alone for methods, results, limitations and the candidate inventory.

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Appendix C. Full fold definitions and family composition

GroupKFold tests all sequences in a held-out family only once over five folds. The table below prints the literal held-out groups from the committed JSON, together with test positivity and the raw fold metrics. It shows how the target population and class balance vary, so the arithmetic mean should not be overread as a pooled population estimate. Fold count is five because that was the frozen implementation, not because five-fold error bounds are known to be stable for 25 unequal families.

C. random

Fold 0: training 1451, testing 363, keyword-positive 108 (29.8%). AUROC 0.97996, AP 0.95185, prevalence baseline AP 0.29752, balanced accuracy 0.91291.

Fold 1: training 1451, testing 363, keyword-positive 108 (29.8%). AUROC 0.98537, AP 0.96454, prevalence baseline AP 0.29752, balanced accuracy 0.93622.

Fold 2: training 1451, testing 363, keyword-positive 109 (30.0%). AUROC 0.97873, AP 0.95845, prevalence baseline AP 0.30028, balanced accuracy 0.93706.

Fold 3: training 1451, testing 363, keyword-positive 109 (30.0%). AUROC 0.98786, AP 0.97860, prevalence baseline AP 0.30028, balanced accuracy 0.94952.

Fold 4: training 1452, testing 362, keyword-positive 108 (29.8%). AUROC 0.97029, AP 0.93431, prevalence baseline AP 0.29834, balanced accuracy 0.89906.

Interpretation: random assignment allows closely related family signatures across folds; this design approximates interpolation under the curated keyword distribution.

C. family held out

Fold 0: training 1306, testing 508, keyword-positive 327 (64.4%). AUROC 0.49617, AP 0.63654, prevalence baseline AP 0.64370, balanced accuracy 0.50153.

Held-out families: three-finger toxin.

Fold 1: training 1459, testing 355, keyword-positive 127 (35.8%). AUROC 0.51274, AP 0.38398, prevalence baseline AP 0.35775, balanced accuracy 0.50174.

Held-out families: phospholipase a2.

Fold 2: training 1497, testing 317, keyword-positive 22 (6.9%). AUROC 0.70216, AP 0.12693, prevalence baseline AP 0.06940, balanced accuracy 0.58451.

Held-out families: crotonamine-myotoxin, disintegrin, ficolin lectin, nucleotide pyrophosphatase/phosphodiesterase, omanin/vespryn, type-b carboxylesterase/lipase, venom metalloproteinase (m12b).

Fold 3: training 1497, testing 317, keyword-positive 46 (14.5%). AUROC 0.56225, AP 0.15401, prevalence baseline AP 0.14511, balanced accuracy 0.47192.

Held-out families: avit (prokineticin), bradykinin-potentiating peptide, crisp, endothelin/sarafotoxin, natriuretic peptide, pdgf/vegf growth factor, peptidase s1, venom complement c3 homolog.

Fold 4: training 1497, testing 317, keyword-positive 20 (6.3%). AUROC 0.68283, AP 0.10508, prevalence baseline AP 0.06309, balanced accuracy 0.60934.

Held-out families: 5'-nucleotidase, flavin monoamine oxidase, multicopper oxidase, ngf-beta, phospholipase b-like, snaclec, true venom lectin, venom kunitz-type.

Interpretation: no named family appears in both train and test for the fold, but phylogenetic and annotation relationships may cross named families. This tests harder transfer, not a clean causal ablation of homology alone.

Methods note: equations, invariants and failure conditions

The equations above are descriptions of implemented calculations, not novel biochemical laws. This appendix connects them to tests and failure boundaries so that counting formula lines does not disguise an unverified

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phenotype. It also explains where the definitions break or cease to be measurable. For exact implementation and unit checks see `venom/scripts/evaluate.py`, `venom/scripts/audit_active_sites.py`, `phosgene/model.py` and the `tests/` folder.

E1. Binary feature map

Binary 3-mer features have no explicit physical state, folding information or catalytic-site geometry. Duplicate exact sequences can violate train/test independence, so exact-sequence groups were collapsed before evaluation. Named family holdout reduces one family-level leakage channel but does not establish independent wet-lab measurements.

E2. Balanced logistic loss

Class weights are estimated on each training fold, and the intercept and coefficients are learned only there. A held-out family can shift both feature support and the relationship between model score and prior probability. The threshold audit exposes this risk: all 120 direct-entry scores lie below 0.5.

E3. AUROC and AP

AUROC is undefined for a test fold with only one observed class; the code marks such folds unassessable rather than using a made-up value. Average precision depends strongly on prevalence, so the baseline AP and positive count accompany each fold. A comparison of the two mean AUROCs is descriptive because test distributions differ.

E4. Attribution site ratio

The X-substitution sensitivity ratio is a quotient of mean absolute logit changes. It would be undefined if mean control delta were zero; the observed denominator is positive, and the measured ratio is 0.67338. Since catalytic active sites are not validated causal features for the Neurotoxin keyword, no mechanistic attribution claim follows.

E5. CAI geometric mean

The codon adaptation index uses log weights and exponentiation, which avoids underflow across the 432-residue coding sequence. A codon with zero reference count cannot be assigned log weight and is rejected rather than silently treated as functional absence. The maximum of 1.0 follows trivially from selecting a maximal count for each amino acid, independent of biology.

E6. Substrate concentration and enzyme units

Dividing mmol per kg meal by liters per kg meal produces mM. One enzyme unit is interpreted as one micromole/min under the assay, converted by 1/1000 into mmol/min. Different assay temperatures, pH, protein secretion and luminal dilution can invalidate that conversion as an in-vivo rate.

E7. Nonnegative integration

The Euler step deducts no more substrate than remains and therefore conserves initial phytate substrate into hydrolyzed plus residual amounts. The mass-balance residual is arithmetic zero by construction, not an independent analytical measurement of product identity or dietary phosphorus fate.

E8. Fecal phosphorus bound

The new-retained quantity cannot exceed baseline fecal phosphorus by the min operation. Thus calculated relative reduction is between zero and one when baseline fecal output is positive. Its numerical value can still be anything in that interval as free fractions vary; this is exactly why matching a literature range by choice of input is not validation.

E9. Beta-binomial posterior

The Beta update assumes exchangeable Bernoulli trials for the chosen narrow endpoint. The five idea builds were selected deliberately and are not exchangeable draws from all project ideas or awards. A posterior on runnable construction cannot be transported to discovery or award odds.

E10. Service and dataset units

The tool denominator is independent scientific services with actually used output, not API calls. The accession denominator is separate direct entry retrievals used in a scored audit, not all IDs in a bulk query. Their units differ: 120 entries from one UniProt service satisfy an entry-count condition while adding one, not 120, to a scientific-service ledger.

14. Reproducibility, limitations and next valid tests

The research record has a defined scope for each outcome, not one pooled pass/fail label. The venom sequence baseline uses reviewed snake-toxin entries, a UniProt keyword and one family ontology supplied by the same database. It is meaningful to report the random versus family-held-out ranking gap and the direct-entry test counts; it would be wrong to transport those numbers to venom potency or antivenom selection. The balanced 120-entry selection creates a 50% positive fraction in each target family by construction, while the original 1,814-entry cohort has 542 keyword positives. AP, threshold sensitivity and any predicted probability should therefore be interpreted against each sample-specific prevalence rather than as a clinical risk score.

The feature map is deliberately simple: contiguous amino-acid 3-mers and logistic weights. Exact-sequence collapse removes identical duplicate rows but not homologs, and the follow-up similarity audit shows why random folds are easier. A family category fed directly to another classifier reaches AUROC 0.87929 on those random folds, which is striking annotation signal but cannot prove which sequence motifs the first classifier used. A genuine independent test would preselect experimentally measured activities, freeze a protein homolog clustering method before outcome inspection, separate training and test by both family and measurement publication, and report calibration as well as discrimination. No such cohort has been obtained in this manuscript. For PhosGene, the AppA reference sequence and codon table are reproducible public records, and translation conservation is an exact software check. CAI 1.0 after choosing the maximally weighted synonym for every amino acid says only that the optimizer follows its reference counts. In an expression experiment the appropriate unit would be paired constructs in a named host with matched promoter, secretion system, culture conditions, protein yield and enzyme activity. A host mismatch, mRNA folding change or secretion failure could reverse the arithmetic ranking. The pig fecal-P endpoint additionally needs feed composition, active enzyme exposure and absorption measurements. No amount of resampling the toy mass-balance equation can replace those original assays.

The 2018 dietary-pig and earlier Enviropig reports are contexts, not exchangeable calibration targets. The former gives diet-specific fecal-P decreases of 45.8% and 34.8% for a multi-enzyme setup; the earlier construct reports up to 75% under different biology. Selecting model inputs to land within these numbers after seeing them would be target leakage. The evidence validator reports ten missing or unsupported input dimensions and a null phenotype for the actual packet. A sensible next test would gather measurements for every dimension in one specified feeding design, assess source relevance before the model is run, and leave any transfer assumption explicit. Even a numerical agreement would then need uncertainty and an independent animal comparison, not just a plausible point estimate.

The award-corpus analysis has a different identification problem. Its 125 rows are awarded project listings, not a registry of all submissions. Titles are a lossy summary of methods, category boundaries vary, and repeated top-award mentions were deduplicated by year/project code. A title-term frequency can guide which ideas to inspect; it cannot say which technique improves award probability. A valid rubric study would recruit independent blinded scorers for the same anonymized idea briefs and compare scores with actual later results, retaining both failures

and ambiguous cases. Neither nonwinner submission records nor such independent rater scores are in the current study.

A shared denominator cannot discharge candidate-specific resource requirements. VenomFunction has 120 direct accession reads that were scored, but only six distinct counted scientific service uses; PhosGene has eleven distinct scientific services and an exploratory 129-entry direct protein-family provenance audit, clearing the narrow accession-count threshold but not an experimentally validated phytase or pig-phenotype gate. Candidate #7 has 120 BioSample records, yet the cohort lacks a multi-hospital interpreted R/S comparison and no genome predictor has been validated. Candidates #3 and #4 now have 120 separately retrieved protein entries for their narrow audits, but not their measured-function or synonymous-expression benchmarks; #5-6 and #8-10 retain source and accession deficits. Repeated endpoint requests, mirrored annotation services, a bulk table with many IDs, and a bibliography of tools are not forty materially used biological services for any one project. These shortfalls remain visible in the ledgers and cannot be repaired by relabeling repositories or supplement pages as evidence.

The paper and companion supplement have complementary purposes. The main text has the full study question, evaluation designs, measured results, failure analysis, ten-build status and limits. The supplement is an audit trail: all 120 toxin entry scores, the 125 listed awards and each attempted external service endpoint. A reader should use the main paper to understand the claims and the supplement to verify individual rows, then inspect the cited JSON/TSV and source URLs for exact sequences, dates and retrieval hashes. The supplement does not add independent experiments, and a long appendix is not a substitute for valid data or methods.

15. Gate-to-evidence decision table

Claim under test	What is directly observed	What remains unproved
Award pattern	125 official awarded C BIO title rows	No nonwinner denominator or award odds
Rubric reliability	Deterministic five-field scoring CLI	No three-person blinded agreement
Venom keyword baseline	1,814 reviewed sequences; fixed random/family folds	No measured potency or independent function cohort
Venom direct test	120 separate entries fetched, checked and scored	Only one service; selected balanced sample
PhosGene optimizer	Translation conserved; 129-entry family scope audit	No expression/secretory yield measurement
Pig phosphorus model	Mass-balance equations and provenance refusal	No source-complete in-vivo input packet
Builds #3-10	Runnable audits and bounded tests in own directories	Separate 120/40 gates and full papers open

The table is a decision aid, not a sum of heterogeneous outcomes. In particular, a false-negative threshold failure and an unmeasured phenotype are different reasons not to ship a scientific predictor. Future work should specify which gate an experiment will answer before collecting more data; adding a service without a biological use cannot close any row.

Candidate	Counted scientific providers	Strict direct-entry status	Principal unresolved test
VenomFunction-2	6/40	120 direct scored UniProt accessions	Independent measured activity and transfer
PhosGene	11/40	129 direct family-audit proteins; no activity assay	Pig-physiology evidence and phenotype
#3 ToxinAudit	5/40	120 direct UniProt entries; 41 target-EC citations	Measured functional negatives
#4 CodonContext	5/40	120 direct proteins; 120 CDS features/one genome	Paired expression constructs
#5 GraphHoldout	3/40	0 individual biological retrievals	Historical source-disjoint prospective edges
#6 TranscriptShift	1/40	0 separate sample files	Comparable independent asthma cohort
#7 MicrobeTransfer	4/40	120 BioSample records, not genome predictor set	Multi-hospital interpreted AST and genomes

#8 PhageHostScope	3/40	97 source phage FASTAs; external 105 incomparable	Compatible external host representation
#9 RNAEvidence	2/40	No individually verified 120-entry ranker set	Independent gene-target assay endpoint
#10 SpatialArtifactGuard	1/40	Bulk matrix and 15,611 aligned spots	Independent site and pixel validation

Counts are per candidate and reflect dated committed ledgers; they may change only when a new material use and its biological interpretation are recorded. The table does not add scores across candidates. A direct BioSample AST record answers a metadata/assay provenance question, not the same question as a genome that is individually fetched and used in a predictor. An accession can be a dataset unit for a well-specified audit while still failing a larger discovery requirement. See `projects/EVIDENCE_LEDGER.md`, `builds/service_ledger.tsv` and `external_tools/ledger.tsv`.

15.1 Interpretation guide for independent readers

A raw score should be paired with its observation unit and a meaningful denominator. The venom random-fold AUROC refers to held-out rows from one curated source with homologous families shared across folds; the held-out-family AUROC refers to a different set of entire families. The two AUROCs are not repeated measurements of the same target population. Likewise, the PhageHostScope AUC refers to pairs held out by trees within one source experiment, while its external *Klebsiella* inventory has no compatible host representation and therefore no external AUC. These distinctions matter more than the size of any single number.

A refusal is a result when the tool has checked the relevant evidence, not a placeholder success. ToxinAudit refuses to label unassayed enzymes negative, PhosGene refuses to infer a pig effect from an in-vitro Km, MicrobeTransfer refuses a hospital-disjoint score from one R/S collector, and GraphHoldout refuses precision when unknown gene-disease pairs are not measured negatives. Each refusal is reproducible code behavior with a stated missing experiment. The next branch of work should target the missing measurement rather than recoding a null outcome as a positive model performance metric.

The current resource counts are audit snapshots. The strict criterion for a counted service is that its output was materially used in the named candidate and the biological interpretation or limitation is logged. An API call that returned a 404, a citation copied into a reference list, and a second website mirroring the same record do not make independent scientific tools. Similarly, accession-wise retrieval and actual use in an analysis differ from spotting many accession strings in a bulk CSV. This distinction is why 120 direct toxin entries count as an entry audit but just one UniProt service.

16. Conclusion and bounded claim

The defensible finding is methodological rather than a new treatment or a measured animal effect. In the protein-keyword task, a random fold looked excellent and a family-disjoint fold fell near chance; direct accession scoring confirmed poor transfer in two selected toxin families. Separate analyses found high train-test sequence resemblance in random folds, substantial family-annotation signal, an unsuccessful threshold transfer, and a negative catalytic-site input-sensitivity proxy. Each is a specific check with its own data and timing. Together they support caution about an easy benchmark, not a universal causal explanation or a clinical classifier.

The AppA branch reached a different endpoint: the coding-sequence optimizer and phosphorus equation are executable and testable, but their available inputs cannot identify a pig fecal-phosphorus reduction. The provenance gate that returns null is a positive software safety property, not a positive biological phenotype. Extending that branch requires original matched expression and feeding data rather than selecting free constants to imitate a published decrease. The award/rubric branch likewise yields a transparent winner-only catalog and scoring instrument, not an estimate of which method wins or agreement among independent judges.

Eight additional candidate builds now broaden the program inventory, but their research units vary: experimental citations, synthetic CDS rules, curated graph edges, GEO cohorts, AST metadata, host-range plates, assay records and spatial spots. These units cannot be exchanged for each other or pooled to satisfy ten separate accession and

service gates. Only the bounded claims in this paper and the source-limited inventory should be communicated as current results. The linked supplement makes the data trail inspectable, while the candidate-specific short reports and repository tests show what is runnable and where each gate still fails. Future successes should add new measurements and independent evaluations without erasing these negative waypoints.

References and record locator

Official award pages: `survey/data/official_cbio_awards.tsv`, per-row official URLs

UniProtKB API documentation: https://www.uniprot.org/help/api_queries

Snake toxin annotation background: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3393831/>

2018 pig feeding experiment, full text: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5963925/>

Original Enviropig experiment: <https://europepmc.org/article/MED/11479566>

AppA in-vitro kinetics: <https://pubmed.ncbi.nlm.nih.gov/10696472/>

Measured AppA mutant comparison: <https://www.frontiersin.org/journals/bioengineering-and-biotechnology/articles/10.3389/fbioe.2023.1167530/full>

Codon harmonization paired-yield feasibility: <https://doi.org/10.1371/journal.pone.0002189>

Specific direct source/tool URLs: `external_tools/ledger.tsv` and linked result JSON

Rubric and prospective protocol: `src/isefdev/rubric.py`; `protocols/prospective_2026-09-25.md`